

Problem Set: Mathematical Foundations

SOLUTIONS

1. (a)

$$\begin{aligned}\mathbf{v} \cdot \mathbf{v} &= 1 \\ \beta^2(9 + 16) &= 1 \\ \beta^2 &= \frac{1}{25} \\ \beta &= \pm \frac{1}{5}\end{aligned}$$

(b) Let:

$$\mathbf{w} = \begin{bmatrix} a \\ b \end{bmatrix}$$

for some $a, b \in \mathbb{R}$

Normality condition:

$$\begin{aligned}\mathbf{w} \cdot \mathbf{w} &= 1 \\ a^2 + b^2 &= 1\end{aligned}$$

Orthogonality condition:

$$\begin{aligned}\mathbf{v} \cdot \mathbf{w} &= 0 \\ \beta(3a + 4b) &= 0\end{aligned}$$

Rearrange 2nd condition and sub in 1st:

$$\begin{aligned}b &= -3a/4 \\ \implies a^2 + \frac{9}{16}a^2 &= 1 \\ \implies \frac{25}{16}a^2 &= 1 \\ \implies a^2 &= \frac{16}{25} \\ \implies a &= \frac{4}{5} \\ \implies b &= \frac{-3}{5}\end{aligned}$$

So:

$$\mathbf{w} = \frac{1}{5} \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

(c) No, because $-\mathbf{w}$ is also orthonormal to $\mathbf{v}$

(d) Yes

(e) If $\mathbf{A}$ is invertible then $\det \mathbf{A} \neq 0$ by the Invertible Matrix Theorem.

$$\begin{aligned} \det \mathbf{A} &= (4.9) - \alpha \cdot \alpha \\ &= 36 - \alpha^2 \\ \implies 36 - \alpha^2 &\neq 0 \\ \implies 36 &\neq \alpha^2 \\ \implies \alpha &\neq \pm 6 \end{aligned}$$

(f) If $\mathbf{A}$ is positive definite then its eigenvalues, $\{\lambda_i\}_{i=1}^2$, are positive.

Eigenvalues are satisfied by the following characteristic polynomial:

$$\begin{aligned} \det(\mathbf{A} - \lambda_i \mathbf{I}) &= 0 \\ \begin{vmatrix} 4 - \lambda_i & \alpha \\ \alpha & 9 - \lambda_i \end{vmatrix} &= 0 \\ (4 - \lambda_i)(9 - \lambda_i) - \alpha^2 &= 0 \\ (\lambda_i)^2 - 13\lambda_i + 36 - \alpha^2 &= 0 \\ \implies \lambda_i &= \frac{13 \pm \sqrt{13^2 - 4(36 - \alpha^2)}}{2} \end{aligned}$$

So for $\lambda_i > 0$:

$$\begin{aligned} 13 - \sqrt{13^2 - 4(36 - \alpha^2)} &> 0 \\ 13^2 &> 13^2 - 4(36 - \alpha^2) \\ 36 &> \alpha^2 \\ \implies -6 &< \alpha < 6 \end{aligned}$$

- (g) If $\mathbf{A}$ represents a covariance matrix then the diagonal elements give the variances, σ_1^2, σ_2^2 :

$$\begin{aligned}\sigma_1^2 = 4 &\implies \sigma_1 = 2 \\ \sigma_2^2 = 9 &\implies \sigma_2 = 3\end{aligned}$$

The correlation, ρ , is given by the off-diagonal elements, so that:

$$\begin{aligned}\rho\sigma_1\sigma_2 &= \alpha \\ \rho &= \frac{\alpha}{\sigma_1\sigma_2} \\ &= \frac{-3}{2 \times 3} \\ &= -0.5\end{aligned}$$

- (h) The shape will be an ellipse

The ellipse will be axis-aligned

The ratio of the semi-major to the semi-minor axis will be: 3:2

2. (a) Note that the form of $f(x)$ is:

$$f(x) = 2x^2 - 2(a_1 + a_2)x + (a_1^2 + a_2^2)$$

So, forming the Hessian:

$$\mathcal{H}(x) = 4$$

Thus the Hessian is positive definite.

So $f(x)$ is strictly convex

So the solution is globally optimal and unique (by properties of strictly convex functions)

- (b) Note that $g(x)$ has a characteristic ‘V’ shape, with change of slope at b (a sketch would also be acceptable in order to demonstrate this)

State that $g(x)$ is convex, but not strictly convex

So the solution is globally optimal (by properties of convex functions).

But the solution is unique (by geometry).

- (c) Note that $h(x)$ has a characteristic ‘_/’ shape, with changes of slope at b_1 and b_2 (a sketch would also be acceptable in order to demonstrate this)

State that $h(x)$ is convex, but not strictly convex

So the solution is globally optimal (by properties of convex functions).

And the solution is not unique (by geometry).

- (d) Note that $k(x) = f(x) + h(x)$, and that the sum of a convex function and a strictly convex function is strictly convex.

So the solution is globally optimal (by properties of convex functions)

And the solution is unique (by properties of strictly convex functions)

3. **Consider** $x \geq 1$:

$$(1 - x) \leq 0 < e^{-x}$$

Consider $0 < x < 1$:

Binomial expansion, valid for $|x| < 1$:

$$\frac{1}{1 - x} = 1 + x + x^2 + x^3 + \dots$$

Taylor expansion of e^x :

$$e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \dots$$

For positive x this implies:

$$\begin{aligned} e^x &< \frac{1}{1 - x} \\ \implies (1 - x) &< e^{-x} \end{aligned}$$

Consider $x = 0$:

$$\implies (1 - x) = 1 = e^0 = e^{-x}$$

Consider $x < 0$:

Let $y = -x > 0$:

$$\implies (1 - x) = (1 + y)$$

And:

$$\implies e^{-x} = e^y = 1 + y + \frac{1}{2}y^2 + \frac{1}{3!}y^3 + \dots$$

So, since $y > 0$:

$$\begin{aligned} (1 + y) &< e^y \\ \implies (1 - x) &< e^{-x} \end{aligned}$$

4. (This solution assumes a linear, risk-neutral, utility function throughout).

(a)

$$\begin{aligned}\mathbb{E}[1 \text{ roll}] &= \frac{1}{6} \times 1 + \frac{1}{6} \times 2 + \frac{1}{6} \times 3 + \frac{1}{6} \times 4 + \frac{1}{6} \times 5 + \frac{1}{6} \times 6 \\ &= \frac{1}{6} \times (1 + 2 + 3 + 4 + 5 + 6) \\ &= 3.5\end{aligned}$$

(b) Assume 1st roll takes place at $t = 0$, 2nd roll takes place at $t = 1$

$$\begin{aligned}\mathbb{E}[2\text{nd roll at } t = 1] &= \frac{1}{6} \times (1 + 2 + 3 + 4 + 5 + 6) \\ &= 3.5\end{aligned}$$

$$\begin{aligned}\mathbb{E}[1\text{st roll at } t = 0] &= \frac{1}{6} \times (\max(1, \mathbb{E}[2\text{nd roll at } t = 1]) + \max(2, \mathbb{E}[2\text{nd roll at } t = 1]) \\ &\quad + \max(3, \mathbb{E}[2\text{nd roll at } t = 1]) + \max(4, \mathbb{E}[2\text{nd roll at } t = 1]) \\ &\quad + \max(5, \mathbb{E}[2\text{nd roll at } t = 1]) + \max(6, \mathbb{E}[2\text{nd roll at } t = 1])) \\ &= \frac{1}{6} \times (3.5 + 3.5 + 3.5 + 4 + 5 + 6) \\ &= 4.25\end{aligned}$$

(c) Assume 1st roll takes place at $t = 0$, 2nd roll takes place at $t = 1$, 3rd roll takes place at $t = 2$

$$\begin{aligned}\mathbb{E}[3\text{rd roll at } t = 2] &= \frac{1}{6} \times (1 + 2 + 3 + 4 + 5 + 6) \\ &= 3.5\end{aligned}$$

$$\begin{aligned}\mathbb{E}[2\text{nd roll at } t = 1] &= \frac{1}{6} \times (\max(1, \mathbb{E}[3\text{rd roll at } t = 2]) + \max(2, \mathbb{E}[3\text{rd roll at } t = 2]) \\ &\quad + \max(3, \mathbb{E}[3\text{rd roll at } t = 2]) + \max(4, \mathbb{E}[3\text{rd roll at } t = 2]) \\ &\quad + \max(5, \mathbb{E}[3\text{rd roll at } t = 2]) + \max(6, \mathbb{E}[3\text{rd roll at } t = 2])) \\ &= \frac{1}{6} \times (3.5 + 3.5 + 3.5 + 4 + 5 + 6) \\ &= 4.25\end{aligned}$$

$$\begin{aligned}\mathbb{E}[1\text{st roll at } t = 0] &= \frac{1}{6} \times (\max(1, \mathbb{E}[2\text{nd roll at } t = 1]) + \max(2, \mathbb{E}[2\text{nd roll at } t = 1]) \\ &\quad + \max(3, \mathbb{E}[2\text{nd roll at } t = 1]) + \max(4, \mathbb{E}[2\text{nd roll at } t = 1]) \\ &\quad + \max(5, \mathbb{E}[2\text{nd roll at } t = 1]) + \max(6, \mathbb{E}[2\text{nd roll at } t = 1])) \\ &= \frac{1}{6} \times (4.25 + 4.25 + 4.25 + 4.25 + 5 + 6) \\ &= 4.67\end{aligned}$$

5. (a) (Univariate) Normal or Gaussian distribution

(b) μ

(c)

$$\begin{aligned}\frac{dp}{dx} &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \times \left(-\frac{2(x-\mu)}{2\sigma^2}\right) \\ &= 0 \\ \implies -\frac{2(x-\mu)}{2\sigma^2} &= 0 \\ \implies x &= \mu\end{aligned}$$

6. (a)

$$\begin{aligned}\mathbb{E}[\mathcal{X}] &= \theta \times 1 + (1 - \theta) \times 0 \\ &= \theta\end{aligned}$$

(b)

$$\begin{aligned}\text{Var}[\mathcal{X}] &= \mathbb{E}[(\mathcal{X} - \mathbb{E}[\mathcal{X}])^2] \\ &= \mathbb{E}[\mathcal{X}^2] - (\mathbb{E}[\mathcal{X}])^2 \\ &= (\theta \times 1^2) + ((1 - \theta) \times 0^2) - \theta^2 \\ &= \theta(1 - \theta)\end{aligned}$$

(c)

$$\begin{aligned}\text{log-lik} &= \ln \prod_{i=1}^n p_{\mathcal{X}_i}(x^{(i)}) \\ &= \ln \theta^{n_+} (1 - \theta)^{n - n_+} \\ &= n_+ \ln \theta + (n - n_+) \ln(1 - \theta)\end{aligned}$$

where n_+ is the number of positive outcomes in the sequence

(d)

$$\begin{aligned}\frac{d\text{log-lik}}{d\theta} &= \frac{n_+}{\theta} - \frac{n - n_+}{(1 - \theta)} = 0 \\ \implies \frac{n_+}{\theta} &= \frac{n - n_+}{(1 - \theta)} \\ \implies n_+(1 - \theta) &= (n - n_+)\theta \\ \implies \theta &= \frac{n_+}{n}\end{aligned}$$

For this sequence: $n = 5$, $n_+ = 3$, so:

$$\theta = \frac{3}{5}$$